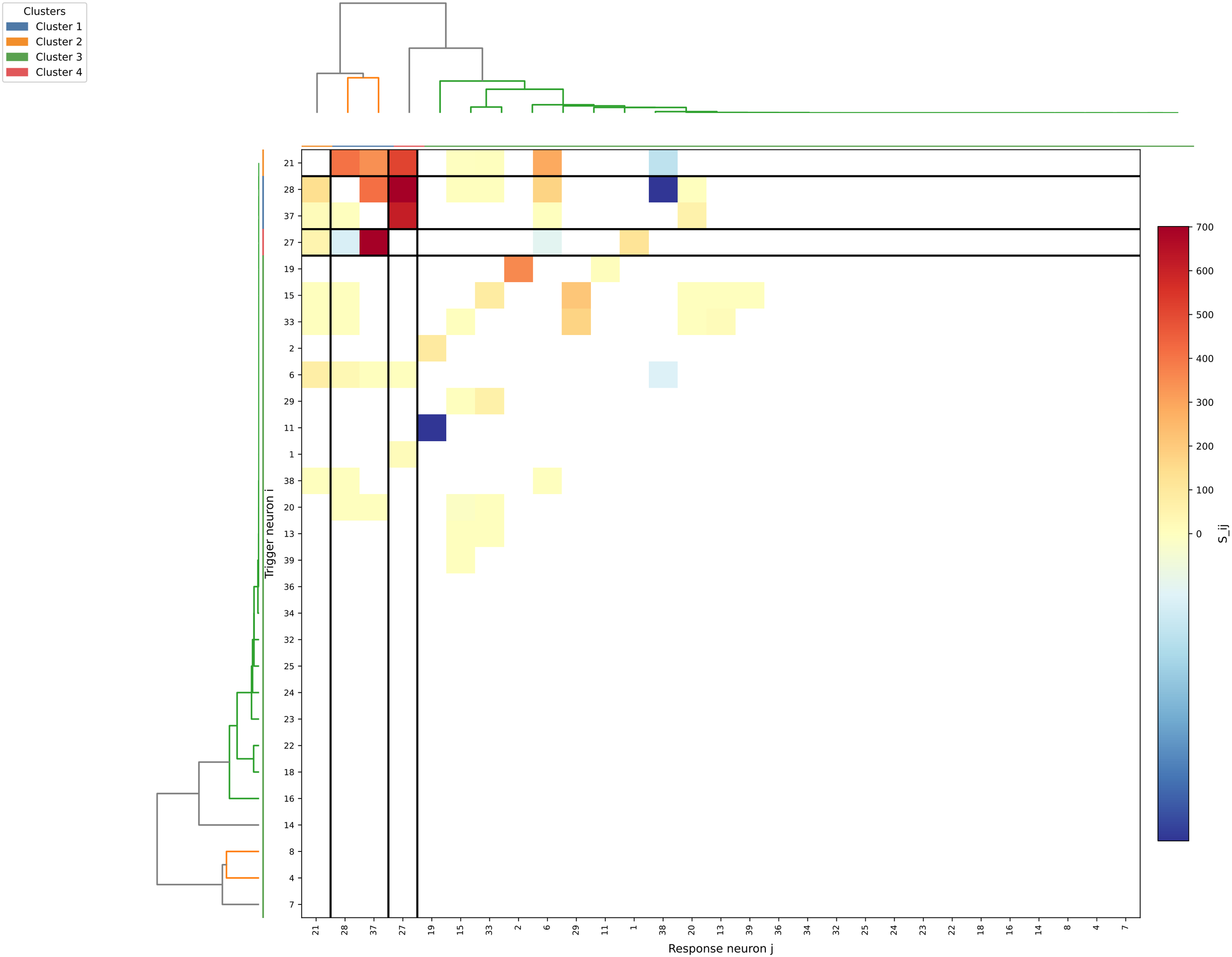S_ij matrix — hierarchical clustering (Ward linkage, k=4)